

Note: Q3.4 on binary trees is not included in this year's exam scope due to curriculum adjustments.

Desk Number _____
Student Number _____
Student Name _____

School of Science and Engineering

FINAL EXAMINATION

Semester 1, 2017

CSC1001 Introduction to Computer Science

Examination Duration: 120 minutes

Reading Time: 10 minutes

This examination has 3 questions.

Exam Conditions:

This is a FORMAL Examination

This is a RESTRICTED OPEN BOOK Exam. Maximum of one (1) sheet of handwritten notes double sided are permitted

Materials Permitted In The Exam Venue:

Maximum of one (1) sheet of handwritten notes double sided are permitted. **NO OTHER MATERIALS PERMITTED**

Any calculators without the functionalities of programming and file storage are permitted.

Materials To Be Supplied To Students:

1 × 12 Page Answer Booklet

Question 1. (15 × 2% = 30%)

Pick the correct option in each of the following sub-questions. Note that only ONE option is correct.

1.1) Binary number **1101111.111** and hexadecimal number **2AC.3** equal to decimal numbers:

- A. **111.875** and **684.1875**
- B. **110.85** and **683.1875**
- C. **111.111** and **684.3**
- D. **110.85** and **683.3**

1.2) Concerning Python language, which of the following statements is incorrect?

- A. Python uses an interpreter to translate source codes into machine codes.
- B. In Python, reserved words cannot be used as variable names.
- C. In Python, a variable has a fixed location in the memory.
- D. **true** and **false** are not reserved words in Python.

1.3) Concerning the following program, which of the following statements is incorrect?

```
if x<2:
    print('Below 2')
elif x<20:
    print('Below 20')
elif x<10:
    print('Below 10')
else:
    print('Something else')
print('Finished')
```

- A. **print('Below 2')** will be executed when $x = 1$.
- B. **print('Below 20')** will be executed when $x = 10$.
- C. **print('Below 10')** will be executed when $x = 8$.
- D. **print('Something else')** will be executed when $x = 20$.

1.4) Concerning the object in Python, which of the following statements is incorrect?

- A. Every object in Python has a unique ID.
- B. The type of an object is determined automatically by Python interpreter according to its value.
- C. Every variable is essentially a reference to an object.
- D. The ID of an object may be changed during the execution of the program.

1.5) Concerning the class in Python, which of the following statements is incorrect?

- A. A class is a template for creating objects in Python.
- B. We can create many different objects from the same class.
- C. In a class, variables and functions are also called data fields and methods.
- D. If two objects are created from the same class, they will always contain identical data fields.

1.6) Concerning the following program, which of the following statements is incorrect?

```
import math

class Circle:
    # Construct a circle object
    def __init__(self, radius = 1):
        self.radius = radius

    def getPerimeter(self):
        return 2 * self.radius * math.pi

    def getArea(self):
        return self.radius * self.radius * math.pi

    def setRadius(self, radius):
        self.radius = radius
```

- A. The purpose of `__init__()` is to initialize the data fields in class `Circle()`.
- B. Class `Circle()` contains only 4 methods.
- C. In method `getPerimeter()`, `self` is a reference pointing to the object from which `getPerimeter()` is invoked.
- D. When creating an object from class `Circle()`, it is acceptable to input 0 arguments.

1.7) Concerning the following program, which of the following statements is incorrect?

```
class A:
    def __init__(self, a, b):
        self.a = a
        self.b = b

    def multiply(self):
        x = a*b
        return x

    def add(self):
        x = a+b
        return x
```

- A. Class `A()` contains two newly defined data fields.
- B. The scope of `self` is throughout the whole body of class `A()`.
- C. Method `add()` will return the sum of two data fields `a` and `b`.
- D. Variable `x` defined in `multiply()` and `add()` are two different local variables.

1.8) Concerning the following two programs, which of the following statements is incorrect?

```
class A:
    def A(self):
        radius = 3
        print(radius)

def main():
    a = A()
    a.A()
```

```
class A:
    def __init__(self):
        radius = 3

    def setRadius(radius):
        self.radius = radius

def main():
    a = A()
    a.setRadius(3)
```

- A. When executing the first program, an error will be triggered.
- B. When executing the second program, an error will be triggered.
- C. The first class `A()` will inherit the initializer from its parent class.
- D. The `a.setRadius(3)` statement of the second program aims to set the radius of `a` to 3.

- 1.9) Concerning private members of a class, which of the following statements is incorrect?
- A. In Python, the names of private data fields and private methods start with 2 underscores and do not end with 2 underscores.
 - B. Private data fields can be accessed within a class.
 - C. Private methods can be directly accessed outside the class definition.
 - D. Private data fields are defined to prevent potential data corruption.
- 1.10) Concerning class inheritance in Python, which of the following statements is incorrect?
- A. Inheritance enables you to define a special class and later extend it to define more general classes.
 - B. A subclass may inherit data fields and methods from its superclass.
 - C. When defining a subclass, the name of its superclass should be placed in the parenthesis after the name of the subclass.
 - D. In Python, every user-defined class is essentially a subclass of other classes.
- 1.11) Concerning tuple, which of the following statement is correct?
- A. A tuple is a sequence of elements which are indexed from 1.
 - B. The content of a tuple can be changed during the runtime.
 - C. Tuple is more efficient compared with list in Python.
 - D. Tuples are comparable and two tuples are equal if their first elements are equal.
- 1.12) Concerning the following program, which of the following statements is incorrect?

```
class Student:
    def __str__(self):
        return 'Student'

    def printStudent(self):
        print(self.__str__())

class GraduateStudent(Student):
    def __str__(self):
        return 'Graduate student'

def main():
    g = GraduateStudent()
    print(g.__str__())

main()
```

- A. The output of this program is 'Student'.
 - B. Class `GraduateStudent()` has inherited the method `printStudent()`.
 - C. Method `__str__()` has been overridden in class `GraduateStudent()`.
 - D. In class `GraduateStudent()`, method `printStudent()` will be inherited from class `Student()`.
- 1.13) Concerning the following program, which of the following statements is correct?

```

class C1:
    def __init__(self):
        self.f = 1

    def output(self):
        print('In C1, the f is:', self.f)

class C2(C1):
    def __init__(self):
        self.f = 2

    def output(self):
        print('In C2, the f is:', self.f)

class C3(C2):
    def __init__(self):
        self.f = 3

class C4(C3):
    def __init__(self):
        self.f = 4

a=C4()
print(a.f)
a.output()

```

- A. On the inheritance chain defined in this program, class **C1()** is the highest class.
- B. In this program, the **output()** method defined in class **C2()** will be executed.
- C. When running this program, an error will be triggered.
- D. The output of this program is:

4
In C2, the f is: 2

- 1.14) Concerning data structure and algorithm, which of the following statements is incorrect?
- A. Data structure concerns how to organize and access data.
 - B. An algorithm is a step-by-step procedure for performing some task in a finite amount of time.
 - C. The quality of an algorithm can be determined by measuring its running time only.
 - D. When analysing the running time of an algorithm, two commonly used approaches are experimental analysis and asymptotic analysis.
- 1.15) Concerning algorithm analysis, which of the following statements is incorrect?
- A. In algorithm analysis, we focus on the growth rate of the running time as a function of the input size.
 - B. The result of asymptotic analysis is platform independent.
 - C. An algorithm with a time complexity of **$O(n)$** will always perform better than an algorithm with a time complexity of **$O(2^n)$** .
 - D. A polynomial time algorithm is usually considered as tractable.

Question 2. (10 × 4% = 40%)

Pick the correct option/s in each of the following sub-questions. Note that there may be MULTIPLE correct options for each sub-question.

- 2.1) Concerning the following program, which of the following statement/s is/are correct?

```
class A:
    def __init__(self, i = 0):
        self.i = i

    def m1(self):
        self.i += 1

class B(A):
    def __init__(self, j = 0):
        super().__init__(3)
        self.j = j

    def m1(self):
        self.i += 1

def main():
    b = B()
    b.m1()
    print(b.i)
    print(b.j)

main() # Call the main function
```

- A. Class **B()** has only one super class.
 B. In class **B()**, the initializer of **A()** is accessed by calling function **super()**.
 C. Class **B()** has only one data field.
 D. When running this program, an error will be triggered.
- 2.2) Concerning the following program, which of the following statement/s is/are correct?

```
a=1
b=2
class A:
    def __init__(self):
        self.a = 10
        self.b = 20
    def B(self):
        a = 100
        print(a)
def main():
    obj = A()
    print(obj.a)
    obj.B()
    print(a)
main()
```

- A. In this program, three different variables **a** have been defined.
 B. Data field **a** of class **A()** can only be accessed within the class definition.
 C. The value of global variable **a** will initially be set as 1, and then changed to 100.
 D. The output of this program is
 10
 100
 100

- 2.3) Concerning the following two programs, which of the following statement/s is/are correct?

```
class Person:
    def getInfo(self):
        return "Person"

    def printPerson(self):
        print(self.getInfo())

class Student(Person):
    def getInfo(self):
        return "Student"

Person().printPerson()
Student().printPerson()
```

```
class Person:
    def __getInfo(self):
        return "Person"

    def printPerson(self):
        print(self.__getInfo())

class Student(Person):
    def __getInfo(self):
        return "Student"

Person().printPerson()
Student().printPerson()
```

- A. The output of the first program is:
 Person
 Student
- B. The output of the second program is:
 Person
 Student
- C. In the first program, the `getInfo()` method will be overridden in class `Student()`.
- D. In the second program, the `__getInfo()` method will be overridden in class `Student()`.
- 2.4) Concerning algorithm analysis, which of the following statement/s is/are correct?
- A. The big-Oh notation allows us to say that a function `f(n)` is always less than or equal to another function `g(n)` up to a constant factor.
- B. The big-Oh notation is usually used to characterize the running time of an algorithm in the asymptotic sense.
- C. Function $5n^4 + 6n^3 + 2n^2 + 2^n + 2n$ is $O(n^4)$.
- D. When we analyse an algorithm, we are usually interested at its average performance regardless of the input size.
- 2.5) Concerning recursion, which of the following statement/s is/are correct?
- A. Recursion is a programming technique by which a function makes one or more calls to itself during execution.
- B. To write a recursion program, we usually need to define the problem in a recursive way first.
- C. If a recursive function is designed so that each invocation of the body makes at most one new recursive call, it is called a linear recursive function.
- D. If a recursive function may make more than one recursive calls, it is called a multiple recursive function.
- 2.6) Concerning the binary search algorithm, which of the following statement/s is/are correct?
- A. The purpose of this algorithm is to find out whether a target element exists in a given sequence of elements.
- B. Binary search algorithm can be applied on an unsorted sequence.
- C. The time complexity of binary search is $O(\log n)$.
- D. Binary search is usually more efficient than sequential search.
- 2.7) Concerning stack, which of the following statement/s is/are correct?
- A. A stack can be accessed based on the "last in first out" principle.
- B. More than one elements of a stack can be accessed simultaneously if necessary.
- C. The time complexity of Inserting an element into a stack is linear.

- D. The time complexity of removing an element from a stack is constant.
- 2.8) Concerning queue, which of the following statement/s is/are correct?
- A. Data are saved sequentially in a queue.
 - B. Data can only be removed from the tail of a queue.
 - C. Data can only be inserted at the head of a queue.
 - D. A queue can only be implemented based on a list.
- 2.9) Concerning linked list, which of the following statement/s is/are correct?
- A. We can identify the tail of a singly linked list as the node having **None** as its next reference.
 - B. The time complexity of inserting a node at the tail of a linked list is constant.
 - C. The time complexity of removing a node at the tail of a linked list is constant.
 - D. Every node in a doubly linked list contains at least two references.
- 2.10) Concerning a binary tree, which of the following statement/s is/are correct?
- A. In a binary tree, data are saved hierarchically.
 - B. In a non-empty binary tree, there is only one node which has no parent.
 - C. Every node on a binary tree contains at least three references.
 - D. In a binary tree, every node has zero or two child nodes.

Question 3. (5% + 5% + 7% + 5% + 8% = 30%)

Answer the following questions.

3.1) Concerning the following program

```
def bubble(bubbleList):
    listLength = len(bubbleList)
    while listLength > 0:
        for i in range(listLength - 1):
            if bubbleList[i] < bubbleList[i+1]:
                buf = bubbleList[i]
                bubbleList[i] = bubbleList[i+1]
                bubbleList[i+1] = buf
        listLength -= 1
    return bubbleList

def main():
    bubbleList = [3, 4, 1, 2, 5, 6, 0, 10, 7]
    print(bubble(bubbleList))
main()
```

Answer the following questions:

- A. What are the outputs of this program?
- B. Which sorting algorithm is applied in this program?
- C. What is the time complexity of this sorting algorithm? How do we prove this?

3.2) Concerning the following program


```
def facFunc(n):
    if n < 0:
        print('Invalid input.')
        return None
    elif n == 0:
        return 1
    else:
        a = n * facFunc(n - 1)
        print(n)
        return a
print(facFunc(5))
```

Answer the following questions:

- A. This program is used to calculate which function?
- B. Write down the corresponding recursive definition of this program.
- C. How many times function `facFunc()` will be called in total and write down the output.

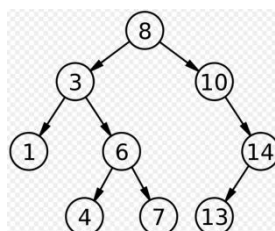
3.3) Write a function which takes two inputs x and n , and outputs x^n . This function should be a recursive function and have a time complexity of $O(\log n)$.

3.4) Concerning the following program, and assume that input t will be a reference pointing to the root of a binary tree. (Out of scope for the 2026 exam)

```
def DFSearch(t):
    if t:
        print(t.element)
        if (t.left is None) and (t.right is None):
            return
    else:
        if t.left is not None:
            DFSearch(t.left)
        if t.right is not None:
            DFSearch(t.right)
```

Answer the following questions:

- A. Which algorithm is implemented in this function?
- B. What is the time complexity of this algorithm?
- C. Is this function a linear recursive or multiple recursive function?
- D. If input t is referencing to the root of the following tree, what would be the outputs of this function?



3.5) Answer the following two questions:

- A. Write a function, which takes a variable t as the input. Input argument t will be a reference pointing to the root of a binary tree. This function should use a Breadth First Search algorithm to print out the values of all nodes on this binary tree.

B. If input t is referencing to the root of the binary tree shown in question 3.4, what would be the outputs of your function?

END OF EXAMINATION